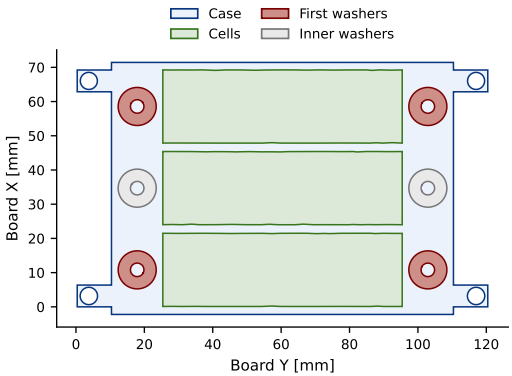
The outer washers meet the case before the cells



Plan view through the cover. Red marks the first four loaded washers.

Lowering the case from 3 mm

The four outer washers touch after 0.603 mm of travel. The rigid tabs are still 2.397 mm from the PCB. The first cell touches after 1.760 mm.

Flush installation

At 0.80 mm travel, the four screw drives carried **80 N with nylon** and **116 N with aluminium**. All four tabs were still clear of the PCB. The through-layer PLA reference was crossed at 0.80–0.95 mm with aluminium and 0.95–1.10 mm with nylon.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

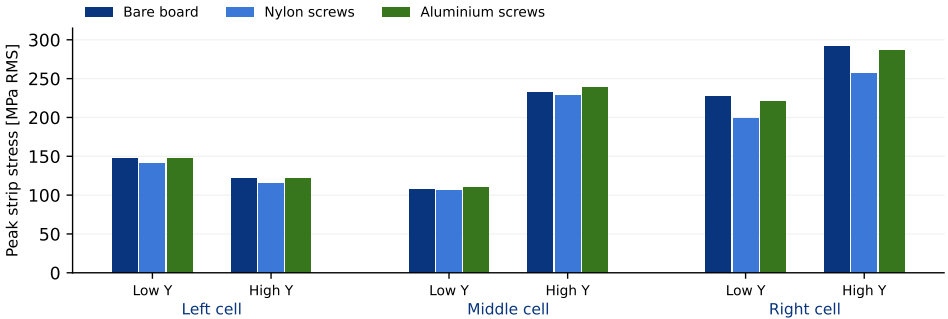
Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

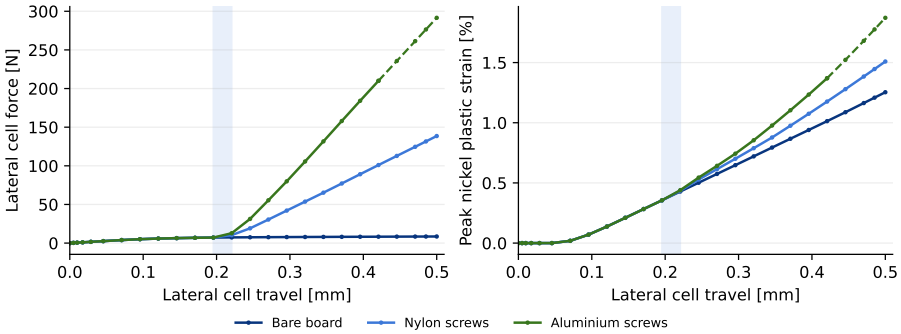
The raised nylon case lowers vibration stress in all six nickel strips



With base X excitation, the highest strip stress was **291.9 MPa RMS** bare, **256.2** with nylon screws and **286.5** with aluminium. All three maxima occurred at the same corner of the right-hand cell attachment.

In the seeded history, all three elastic models crossed the local pull study's assumed 185 MPa yield stress within 39 ms. Both raised cases still had clearance at the cells. With base Z, the largest lengthwise force at a right-cell attachment rose from 2.07 N RMS bare to 4.14 N with nylon and 3.55 N with aluminium.

Both raised cases limit cell travel under the same pull

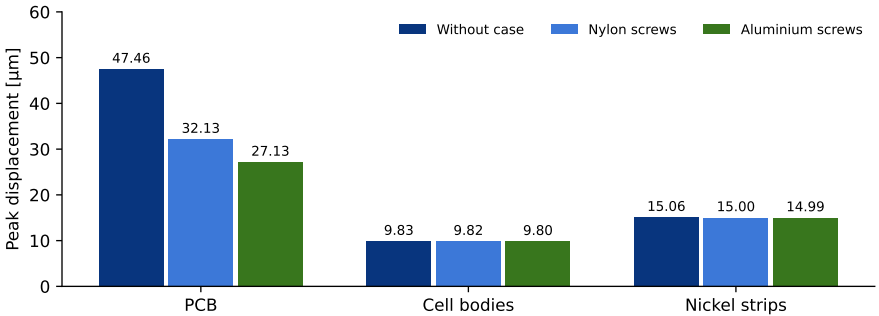


At **8.54 N**, the bare cell travelled 0.50 mm. Both raised cases travelled about 0.20 mm. Their difference was smaller than the spacing between saved states.

At the same **0.421 mm travel**, the required pull was 8.26 N bare, 101.0 N with nylon screws and 210.0 N with aluminium. The aluminium case crossed the first printed-PLA tensile reference in the next increment.

Positive X pull at 3 mm stand-off. Fixed screw tops and rigid tab bearings transmit a couple at every screw.

Aluminium screws further reduced PCB deflection during tightening

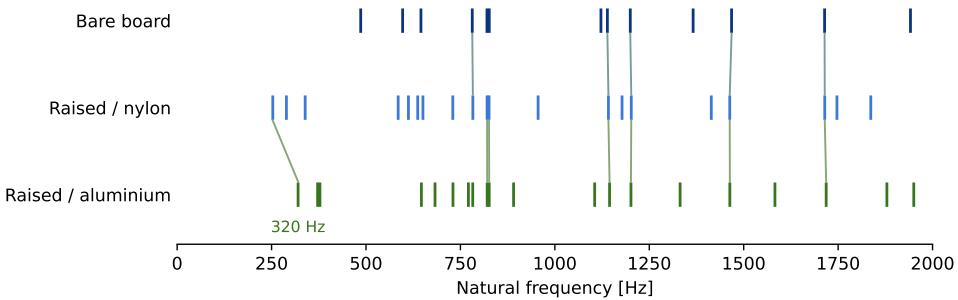


The same PCB node moved 47.46 μm bare, 32.13 μm with nylon screws and **27.13 μm** with aluminium screws. The reductions from bare were 32% and 43%, respectively.

With the raised mounts, all cell, strip and washer gaps stayed open. During the separate flush nylon installation, outer-washer contact pulled a PCB corner **0.328 mm toward the cells** by 0.95 mm of screw travel.

Bars show the six original bolts locked after tightening to 750 N each. Both cased bars use 3 mm stand-offs.

Both raised cases introduce modes below 486 Hz

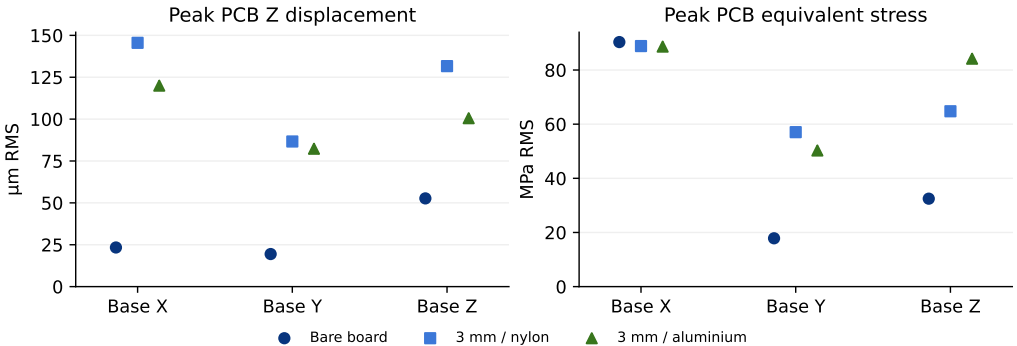


The first cased mode rises from **253 Hz with nylon** to **320 Hz with aluminium**. Its shape remains closely matched across 399,194 solid nodes (MAC 0.989).

The closest match to bare mode 20 (2364 Hz) is mode 27 in both raised assemblies, at 2365 Hz with nylon and 2368 Hz with aluminium. With the original unpreloaded contacts, mode 1 stayed at 66.61 Hz in all three assemblies.

Bolt-locked flexible assemblies at 3 mm stand-off. Undamped mode shapes about the tightened state.

Both raised mounts increase board-normal motion under each base input



With base Z excitation, peak PCB stress rose from **32.5 MPa RMS** bare to **64.8** with nylon screws and **84.2** with aluminium. Aluminium reduced board travel relative to nylon in all three tests.

Under base X, peak PCB stress fell by less than 2% for both mounts. The spectra use the locked, open-gap contact state. The seeded X record predicts cell contact for both mounts during the full nine seconds.